

MOUNT SHASTA, CA, USA

WMO: 725957

Lat: 41.333N

Lon: 122.333W

Elev: 1078

StdP: 89.03

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24215

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -7.7	(c) -5.7	(d) -14.0	(e) 1.3	(f) -1.6	(g) -11.4	(h) 1.6	(i) -0.9	(j) 6.0	(k) 3.7	(l) 5.4	(m) 3.4	(n) 0.8	(o) 50	(p) 0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	33.3	17.9	31.9	17.3	30.2	16.5	19.8	30.1	18.7	29.3	17.6	28.3	1.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	13.1	25.7	14.1	11.4	24.5	12.8	10.5	23.4	60.9	30.0	56.8	29.3	53.5	28.2	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.3	4.7	4.1	-11.8	36.2	2.8	1.5	-13.8	37.3	-15.5	38.2	-17.1	39.0	-19.2	40.1
			-12.5	21.5	2.7	1.7	-14.5	22.7	-16.1	23.7	-17.6	24.7	-19.6	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	2.6	3.6	5.8	8.3	12.8	16.9	20.8	19.8	16.5	10.7	5.3	2.0
	DBStd	7.41	3.24	3.27	3.64	4.04	4.13	3.90	2.80	2.64	3.52	3.59	3.56	3.29
	HDD10.0	1082	230	181	138	80	20	3	0	0	2	35	146	248
	HDD18.3	3072	488	414	390	301	176	71	9	14	73	238	391	506
	CDD10.0	1248	0	1	7	29	107	208	334	304	197	56	4	0
	CDD18.3	194	0	0	0	0	4	27	86	60	18	1	0	0
	CDH23.3	3998	0	0	1	24	174	536	1467	1203	548	45	0	0
	CDH26.7	1534	0	0	0	2	42	173	642	488	179	7	0	0
Wind	WSAvg	1.4	1.4	1.5	1.6	1.7	1.6	1.6	1.1	1.0	1.2	1.3	1.3	1.5
Precipitation	PrecAvg	1030	177	152	135	73	47	26	8	11	18	56	142	179
	PrecMax	1909	698	554	479	230	236	96	43	65	98	247	437	659
	PrecMin	361	5	5	11	2	0	0	0	0	0	0	13	2
	PrecStd	329	134	125	109	56	51	26	11	14	23	57	106	147
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.0	21.7	26.2	30.5	33.3	36.0	35.1	32.8	27.4	19.8	12.7
		MCWB	7.0	9.0	10.9	13.0	15.7	18.1	19.1	17.9	16.5	14.9	11.1	6.9
	2%	DB	12.2	14.8	18.8	23.1	27.6	30.7	33.8	32.7	30.7	24.1	17.1	10.3
		MCWB	6.0	7.7	9.5	12.0	15.3	17.2	18.4	17.5	15.4	13.0	9.8	6.0
	5%	DB	9.8	12.0	16.3	20.4	25.2	28.6	32.2	31.2	28.8	22.1	14.0	8.8
		MCWB	5.4	6.2	8.2	10.7	13.6	16.3	17.7	16.8	15.0	12.0	8.1	5.7
	10%	DB	7.8	9.4	13.3	17.4	22.6	26.6	30.5	29.6	27.0	19.2	11.7	7.4
		MCWB	4.9	5.3	7.1	9.6	12.7	15.5	17.2	16.1	14.7	10.8	7.0	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.1	12.1	14.3	17.8	20.4	21.7	20.9	18.9	15.4	12.3	8.9
		MCDB	11.7	16.1	19.9	23.6	27.4	29.8	31.8	30.2	29.5	24.5	17.8	10.6
	2%	WB	7.4	8.5	10.3	12.6	15.9	18.6	20.2	19.0	17.0	14.0	10.6	7.5
		MCDB	10.1	13.1	16.9	21.6	25.8	28.7	30.3	29.8	27.5	22.8	15.3	9.1
	5%	WB	6.3	7.1	9.0	11.2	14.6	17.2	19.0	17.9	15.8	12.6	9.1	6.5
		MCDB	9.0	10.8	14.9	19.2	23.9	27.0	29.7	28.8	26.5	20.5	12.9	8.2
	10%	WB	5.1	5.8	7.7	9.8	13.2	15.9	18.0	16.9	14.9	11.3	7.8	5.1
		MCDB	7.6	9.1	12.5	16.4	21.5	25.5	28.7	27.9	25.7	18.2	10.7	7.0
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.5	11.0	12.7	14.4	15.7	18.4	18.9	17.8	14.5	10.3	8.0
		MCDBR	12.9	15.0	17.2	19.3	19.4	19.1	20.5	20.8	21.5	19.5	15.1	11.2
		MCWBR	8.0	9.0	9.6	10.0	9.1	8.8	8.8	9.0	9.7	10.1	9.3	7.5
	5% WB	MCDBR	9.9	12.5	15.2	17.6	17.9	17.9	18.8	19.2	19.5	17.3	13.3	8.5
		MCWBR	6.7	8.0	8.9	9.6	8.9	9.3	9.1	9.2	9.5	9.4	8.5	6.3
Clear-Sky Solar Irradiance	taub		0.279	0.284	0.289	0.305	0.315	0.308	0.310	0.306	0.293	0.291	0.276	0.281
	taud		2.553	2.563	2.579	2.529	2.536	2.590	2.596	2.590	2.611	2.618	2.623	2.550
	Ebn at Noon		892	941	970	969	959	963	958	956	953	920	894	858
	Edh at Noon		69	80	88	100	102	97	95	93	84	75	64	65
All-Sky Solar Radiation	RadAvg		1.95	2.87	4.04	5.53	6.87	7.95	8.10	7.19	5.70	3.79	2.25	1.62
	RadStd		0.32	0.36	0.45	0.55	0.53	0.44	0.29	0.29	0.26	0.35	0.26	0.32

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.41	N/A	N/A	+1.21	N/A	N/A	N/A	N/A	+128	+48
(47) Regional (15 neighbors)		N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	+94	+15

Nomenclature: See separate page